

# Output Ontology (OO)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: CRISISLTLSUM | Context: Mexico City and Central America Bilingual Crisis Journalism

**Output Ontology definition:** Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

## Justification

The output ontology consists of (a) a binary timeline-membership label governed by three conditions — relevant, informative, non-repetitive [Q58] — and (b) a four-axis summary quality rubric (Coherence, Accuracy, Coverage, Overall Quality) rated 1–5 [Q106, Q151]. The three relevance conditions are conceptually generalizable, but were validated only on US English tweets by Anglophone annotators [Q61], and cross-lingual transfer validity to Mexican Spanish tweets covering different event types is unestablished. Critically, the Coherence rubric is explicitly anchored to English: 'A summary is coherent if when read by itself, it's easy to understand and free of English errors' [Q153, Q154]. This makes the coherence axis structurally invalid for Spanish-language output. Additionally, the informativeness criterion encodes US-centric judgments about what constitutes a meaningful crisis update; the Mexican context includes documented genres (community solidarity posts, citizen verification posts like #Verificado19S) that have distinct informativeness profiles [WEB-23, WEB-24]. No category captures editorial dimensions (source attribution to CENAPRED/SSN/SASMEX, casualty framing, AP Español alignment) that are central to the target deployment.

| ID     | Evidence                                                                                                                                                                                                                                                                                                                                                                                                           |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q58    | "A tweet is labeled as part of the timeline only if it satisfies all the following three conditions: relevant: talks about the same event indicated in the seed tweet, informative: provides facts about the event but not only contains personal points of view, not repetitive: brings in new information." (p.5)                                                                                                |
| Q61    | "First, we use location restriction (QC1) to limit the pool of workers to countries where native English speakers are most likely to be found." (p.5)                                                                                                                                                                                                                                                              |
| Q106   | "Each summary was evaluated by five different workers on a scale from 1 to 5 across four axes: Coherence, Accuracy, Coverage, and Overall quality, as was done by Lai et al. (2022)." (p.8)                                                                                                                                                                                                                        |
| Q151   | "During this task, you will be reading a timeline that consists of a sequence of tweets describing an event and eight different summaries for the event in the timeline. You will rate the quality of each of the eight summaries by four axes: coherence, accuracy, coverage, and overall quality." (p.21)                                                                                                        |
| Q153   | "For each summary, answer the question 'how coherent is the summary on its own?' on a scale 1 to 5. A summary is coherent if when read by itself, it's easy to understand and free of English errors. A summary is not coherent if it's difficult to understand what the summary is trying to say. Generally, it's more important that the summary is understandable than it being free of grammar errors." (p.21) |
| Q154   | "Score of 1: The summary is impossible to understand. Score of 2: The summary has many mistakes or confusing phrasing. Score of 3: The summary has some mistakes or confusing phrasing that made it hard to understand. Score of 4: The summary is mostly clear with only a few minor mistakes or confusing phrasing. Score of 5: The summary is perfectly clear." (p.21)                                          |
| WEB-23 | #Verificado19S citizen-verification genre would be judged differently by Mexican journalists than US MTurk workers ( <a href="https://arxiv.org/abs/2007.05848">https://arxiv.org/abs/2007.05848</a> )                                                                                                                                                                                                             |
| WEB-24 | <a href="https://link.springer.com/article/10.1007/s00779-020-01411-5">https://link.springer.com/article/10.1007/s00779-020-01411-5</a>                                                                                                                                                                                                                                                                            |

## Strengths

- The three-criterion relevance/informativeness/non-redundancy schema [Q58] is conceptually portable and provides a transferable starting point for Spanish-language label adaptation.
- The four-axis summary rubric (Coherence, Accuracy, Coverage, Overall) [Q106] decomposes summary quality in a way that supports targeted adaptation — Accuracy and Coverage axes [Q155, Q157] are linguistically more neutral than

Coherence.

| ID                   | Evidence                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">Q58</a>  | "A tweet is labeled as part of the timeline only if it satisfies all the following three conditions: relevant: talks about the same event indicated in the seed tweet, informative: provides facts about the event but not only contains personal points of view, not repetitive: brings in new information." (p.5)                                                                                                                          |
| <a href="#">Q106</a> | "Each summary was evaluated by five different workers on a scale from 1 to 5 across four axes: Coherence, Accuracy, Coverage, and Overall quality, as was done by Lai et al. (2022)." (p.8)                                                                                                                                                                                                                                                  |
| <a href="#">Q155</a> | "For each summary, answer the question "Does the factual information in the summary accurately match the information in the timeline?" on a scale of 1 to 5. A summary is accurate if it doesn't say things that aren't in the timeline, it doesn't mix up information, and generally is not misleading." (p.21)                                                                                                                             |
| <a href="#">Q157</a> | "For each summary, answer the question "How well does the summary cover the important information in the timeline?" on a scale of 1 to 5. A good summary has good coverage if it mentions the main information from the timeline that's important to understand the event. A summary has poor coverage if someone reading only the summary would be missing several important pieces of information about the event in the timeline." (p.21) |

## Checklist

### Checklist item definitions:

| ID                   | Assessment Question                                          |
|----------------------|--------------------------------------------------------------|
| <a href="#">OO-1</a> | Review output label categories for regional relevance        |
| <a href="#">OO-2</a> | Identify missing regionally specific categories              |
| <a href="#">OO-3</a> | Flag categories encoding non-regional values                 |
| <a href="#">OO-4</a> | Consider stakeholder-driven taxonomy redesign                |
| <a href="#">OO-5</a> | Document taxonomy issues harming structural/content validity |

### Checklist responses:

**OO-1:** The binary timeline-membership label and four-axis quality rubric are partially relevant, but the Coherence axis is explicitly English-anchored [[Q153](#)] and the relevance/informativeness conditions [[Q58](#)] were validated only on US tweets.

**OO-2:** Missing categories include: source-attribution quality (whether CENAPRED/SSN/CNPC are cited correctly), AP Español register conformance, verification-tweet detection (a specific genre in #19S contexts) [[WEB-23](#)], and seismic/volcanic alert-vocabulary correctness (semáforo phases, SASMEX hashtag usage) [[WEB-7](#), [WEB-20](#)].

**OO-3:** The Coherence axis encodes 'free of English errors' [[Q153](#)] as the criterion, which directly encodes Anglophone language norms. The informativeness criterion [[Q58](#)] implicitly encodes US news-reading conventions about what counts as a 'fact' versus 'personal point of view,' and was validated only by Anglophone annotators [[Q61](#)].

**OO-4:** Significant misalignment exists. Stakeholder-driven taxonomy redesign is warranted: a Spanish-language coherence rubric, a register/style axis aligned with AP Español, and addition of source-attribution and verification-correctness axes are needed.

**OO-5:** Structural validity is compromised by the English-anchored coherence rubric; content validity is compromised by missing axes for register, attribution, and verification quality. External validity transfer is therefore not supported.

| ID                   | Evidence                                                                                                                                                                                                                                                                                                            |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
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| <a href="#">Q61</a>  | "First, we use location restriction (QC1) to limit the pool of workers to countries where native English speakers are most likely to be found." (p.5)                                                                                                                                                               |
| <a href="#">Q106</a> | "Each summary was evaluated by five different workers on a scale from 1 to 5 across four axes: Coherence, Accuracy, Coverage, and Overall quality, as was done by Lai et al. (2022)." (p.8)                                                                                                                         |

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| <b>WEB-7</b>  | SASMEX vocabulary defines a specific seed-tweet/follow-up structure not addressed by benchmark output categories ( <a href="https://en.wikipedia.org/wiki/Mexican_Seismic_Alert_System">https://en.wikipedia.org/wiki/Mexican_Seismic_Alert_System</a> )                                                                                                                                                                                     |
| <b>WEB-20</b> | CENAPRED semáforo phase vocabulary is essential output-ontology content for volcanic crises ( <a href="https://www.cenapred.unam.mx/reportesVolcanesMX/">https://www.cenapred.unam.mx/reportesVolcanesMX/</a> )                                                                                                                                                                                                                              |
| <b>WEB-23</b> | #Verificado19S citizen-verification genre would be judged differently by Mexican journalists than US MTurk workers ( <a href="https://arxiv.org/abs/2007.05848">https://arxiv.org/abs/2007.05848</a> )                                                                                                                                                                                                                                       |

## Information Gaps

- AP Español operationalizable criteria not openly accessible — flagged for expert elicitation
- Whether Mexican journalists' relevance judgments diverge systematically from the benchmark's three-criterion definition is empirically unverified

## Requires Expert Verification

- Mexican journalist review of relevance/informativeness boundaries on Spanish crisis tweets
- AP Español editor input on register and source-attribution rubric

## Remediation

**Gap 1: Coherence rubric is explicitly anchored to English [Q153]; missing axes for source attribution, AP Español register, and verification-tweet detection**

**Recommendation:** Construct a Spanish coherence rubric referencing Spanish-language writing norms; add an AP Español-aligned register axis (requires expert elicitation from AP Español editors) and a source-attribution axis covering CENAPRED/SSN/CNPC/SASMEX [WEB-1, WEB-7]; add a verification-quality axis informed by the #Verificado19S model [WEB-23].

| ID            | Evidence                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Q153</b>   | "For each summary, answer the question "how coherent is the summary on its own?" on a scale 1 to 5. A summary is coherent if when read by itself, it's easy to understand and free of English errors. A summary is not coherent if it's difficult to understand what the summary is trying to say. Generally, it's more important that the summary is understandable than it being free of grammar errors." (p.21) |
| <b>WEB-1</b>  | CENAPRED is the authoritative Mexican source for crisis attribution — distinct from US norms ( <a href="https://en.wikipedia.org/wiki/CENAPRED">https://en.wikipedia.org/wiki/CENAPRED</a> )                                                                                                                                                                                                                       |
| <b>WEB-7</b>  | SASMEX vocabulary defines a specific seed-tweet/follow-up structure not addressed by benchmark output categories ( <a href="https://en.wikipedia.org/wiki/Mexican_Seismic_Alert_System">https://en.wikipedia.org/wiki/Mexican_Seismic_Alert_System</a> )                                                                                                                                                           |
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